Adaptive Architecture Quantum Meta-Learning for Bayesian Binary Neural Networks

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Abstract

In this paper, "global optimum"/"global optimum' " is used as shorthand for "global optimum or a tolerance criterion was met" , so this means, for either (i) the true global minimizer or (ii) any point that satisfies the stopping tolerance criterion. The algorithm does not attempt to distinguish between these cases.The well known Nash Embedding Theorem (NET), states that any n-dimensio$n$nal Riemannian manifold can be isometrically embedded in Euclidean space of dimension $n(n+1)(3n+11)/2$, relies fundamentally on the Nash-Moser iteration to resolve the inherent nonlinear ill-posedness of the underdetermined system of partial differential equations (PDEs) induced by the manifold's positive-definite metric tensor. Inspired by the structural logic of this iterative paradigm and quantum variational optimization, we introduce a Unitary Homotopy framework for Quantum Meta learning. While the Nash-Moser iteration is a cornerstone of nonlinear analysis, its application to quantum heuristic design remains an open area; this paper provides the first formal exploration of this cross-disciplinary link We demonstrate that the injection of parameterized topological noise, governed by our 'Iteration Time Stuck' (ITS) metric, functions as a finite-dimensional analogue to the Nash-Moser smoothing operator. Our method allows the Born machine to traverse local minima,or barren plateaus, and coherently revert the topology through unitary inversion, providing a rigorous, regularized refinement framework for complex, highly-correlated parameter spaces. We are using the Nash-Moser concept as a theoretical framework for our heuristics. That said we give proved theroem in the appendix A that support our heuristic link to some dgree.

Using well known theorem in differential near-term quantum circuits,Born machines) can represent complex discrete probability distributions that are difficult to sample classically. Prior work by Nikoloska and Simeone in [1] demonstrated that Born machines can serve as implicit variational posteriors for Bayesian binary neural networks within a meta-learning framework. However, their approach relies on a fixed hardware-efficient ansatz and classical Monte-Carlo sampling, which can suffer from barren plateaus and slow convergence.

We introduce two new algorithmic variants that preserve the original Bayesian meta-learning structure while addressing these limitations. The first variant augments the Born machine with dynamic entanglement adaptation, triggered by an "Iteration-Time-Stuck" (ITS) metric, which inserts parameterized pSWAP gates to modify the circuit topology when optimization stalls. The second variant adds coherent reversibility, using inverse pSWAP operations to restore the

original architecture after escaping a local minimum. Both variants replace classical sampling with Quantum Amplitude Estimation (QAE) to compute expectation values with quadratic precision improvement. Crucially, architecture adaptation and QAE are executed in strictly separated phases, ensuring that amplitude estimation operates on a static, coherent circuit. These mechanisms jointly improve exploration, avoid local minima, and accelerate convergence in variational quantum meta-learning. Generalised macroscopic smoothing and dynamic scheduling protocols are discussed.

While Nash-Moser theory was developed to resolve ill-posedness in infinite-dimensional nonlinear analysis, our algorithm exhibits a formal correspondence to this paradigm. In Nash-Moser, 'loss of derivatives' is mitigated by applying a smoothing operator, solving, and subsequently desmoothing;a proces that is non-destructively inverted to preserve the solution. Our ITS-driven unitary homotopy U_θ functions as a finite-dimensional analogue to this smoothing operator. By dynamically warping the loss landscape to traverse local minima (barren plateaus) and coherently restoring the original operator through unitary inversion U_θ^{-1}, we mirror the Nash-Moser iterative logic of 'deform-solve-revert.' To be more precise in the global Nash embedding it takes a smooth (in the sense of of the corrugation device used by Nash) Riemannian manfold and embeds it into higher dimensional Euclidean space by making the manifold progressively more crumpled. In our thereom "we take the opposite of the above idea and use it in reverse periodically, that is we take uncrumple a geometrical object and unfold it by embedding it into a Lie manifold then we fold it back and we do this peridically until a solution for findign a solution to the qunatum optimization prolblem. The controlling derivative loss. In the Nash-Moser scheme deals with a "missing"/loss of derivates per iteration in other words the Nash-Moser deals with these "missing" derivatives. In our scheme there is also "missing" derivatives but they are "missing" in a different way, but still "missing", our derivaties are "missing" in the sense of "vanishing directional derivatives" in the "barren plateau" problem; when we write about loss of derivatives for the contents of our paper (Theorems etc.) we mean in the "missing" sense above in the connection to the "barren plateau" problem. The above structural parallels suggests that our approach is not merely a heuristic, but a regularized refinement framework appropriate for complex, highly-correlated parameter spaces.

1 On Foundational Priority

Trilogy level Foundational Priority Declaration. The three companion papers (i) A Reversible Smoothing Framework for Degenerate Riemannian Metrics, (ii) A Geometric Nash–Moser Implicit Function Theorem for PDEs, and (iii) Adaptive Architecture Quantum Meta-learning via Unitary Homotopy, collectively establish the complete architectural, geometric, analytic, and algorithmic foundations of the Reversible Nash–Moser / Unitary Homotopy paradigm.

Across this trilogy, we introduce the core mechanisms of the framework—including but not limited to reversible Lie-group smoothing flows, auxiliary metric inflation, bouquet-of-spheres topology, Lie-algebra smoothing, the Geometric Nash–Moser IFT, the Reversible Implicit Function Theorem (RIFT), the Quantum Nash–Moser Escape Principle, ITS-triggered reversible pSWAP deformations, macroscopic smoothing operators, and the reversible deform–solve–revert cycle—together with all derivative extensions of these mechanisms. This foundational claim explicitly encompasses the extension of these mechanisms to resolve landscapes containing multiple global minima and finite attractor sets, and applies universally to general parameterized quantum circuits, including all Variational Quantum Algorithms (VQAs) beyond standard Born machines.

These three papers jointly define the foundational structure of the reversible smoothing paradigm in geometry, PDE theory, and quantum variational optimization. Subsequent works may refine proofs, extend special cases, or introduce minor technical variations, but such developments are derivative elaborations of the unified framework established in this trilogy.

This paper introduces the complete architectural, algorithmic, and conceptual framework for Unitary Homotopy–based quantum meta-learning, including but not limited to the ITS triggered reversible pSWAP deformation, dynamic entanglement adaptation protocols, macroscopic smoothing operators, the Quantum Nash–Moser Escape Principle, the reversible deform–solve–revert cycle, and all derivative extensions of these mechanisms.

Subsequent works may refine proofs, extend special cases, or introduce minor technical variations, but such developments are derivative elaborations of the framework established in this manuscript. This paper therefore serves as the primary reference for the Unitary Homotopy architecture and its reversible smoothing methodology in quantum variational optimization.

2 INTRODUCTION

Bayesian neural networks encode epistemic uncertainty by maintaining a distribution over model parameters. When parameters are binary, representing this distribution becomes challenging. Born machines quantum circuits whose measurement statistics define a probability distribution that offer a natural implicit variational family for such discrete spaces.

Nikoloska and Simeone in [1] showed that a Born machine can model the variational posterior $q_\phi(\theta)$ of binary weights and can be trained via gradient-based meta-learning. Their method uses:

- a static hardware-efficient ansatz,
- Monte-Carlo sampling to estimate expectations,
- density-ratio estimation to approximate KL terms.

However, static ansätze can become trapped in local minima, and classical sampling limits precision. We propose two new variants that retain the original meta-learning objective but introduce:

- Dynamic entanglement adaptation via parameterized pSWAP gates.
- A reversible exploration mechanism using inverse unitaries.
- Quantum Amplitude Estimation for efficient expectation computation.

These modifications preserve implicit distribution of the Born machine while improving optimization dynamics and sample efficiency In all our algorithms in this paper apply to local minima and barren plateaus or actually all known types of traps (e.g. degenerate metric regions or ill-conditioned saddles) because the stagnation detection is not dependant on the type of trap. In practice, escape from a nonglobal local minimum or a flattened barren plateau region occurs precisely when the ITS criterion detects stagnation, at which point the reversible pSWAP based deformation is activated to lift the parameters into an auxiliary manifold before coherently reverting the topology..In this paper, "global optimum" is used as shorthand for either (i) the true global minimizer or (ii) any point that satisfies the stopping tolerance criterion. The algorithm does not attempt to distinguish between these cases. When we mean converges to a global optimium in this paper we mean it in the sense of either (i) or (ii) above-so global covergence may mean it stops because of a tolerance criterion.

3 VARIANT 1: BASE THREE-PART ALGORITHM

3.1 Variational Quantum Optimization (Base Loop)

As in the original paper, each task τ optimizes variational parameters ϕ_τ by minimizing the free energy:

$$F(\phi_\tau) = -E_{q_{\phi_\tau}}[\log p(D_\tau|\theta)] + \beta\,\mathrm{KL}(q_{\phi_\tau}(\theta)\|p(\theta)). \tag{1}$$

The inner-loop update is:

$$\phi_\tau \leftarrow \xi - \eta\nabla_\xi F(\xi), \tag{2}$$

where ξ is the meta-learned initialization.

3.2 Dynamic Architecture Search via ITS-Triggered pSWAP Gates

Static PQCs often stagnate due to barren plateaus or poor entanglement structure. We introduce an Iteration-Time-Stuck (ITS) counter:

- If $F(\phi_\tau)$ does not improve for s iterations, ITS increases.
- When ITS exceeds a threshold, the circuit enters a Jitter Phase.

pSWAP Injection

A parameterized SWAP gate is inserted between qubits i and $i+1$: $\text{pSWAP}_{i,i+1}(\theta_{ITS})$. The global unitary is implicitly defined by inserting the 2-qubit gate into the circuit:

$$U_{jitter} = I^{\otimes(i-1)} \otimes \text{pSWAP}_{i,i+1}(\theta_{ITS}) \otimes I^{\otimes(n-i-1)}. \tag{3}$$

Increasing $\theta_{ITS} \propto ITS$ perturbs the entanglement pattern, changing the Born machine's probability landscape. Once the loss decreases, ITS resets and the pSWAP parameter returns to zero. This provides a structural exploration mechanism unavailable in classical VI.

3.3 Quantum Amplitude Estimation (QAE) for Expectation Values

The original method estimates expectations using K Born-machine samples. We replace this with QAE,which can be thought of as a variation of Grover's algorithm to compute a mean, which achieves error $O(1/K)$ vs $O(1/\sqrt{K})$. This gives a faster way to compute the mean expectation that given in [1] which uses a density ratio estimation. To apply QAE, we define an oracle that encodes:

$$f(\theta) = \log p(D_\tau|\theta) + \beta \log \frac{q_{\phi_\tau}(\theta)}{p(\theta)}. \tag{4}$$

Architecture adaptation (pSWAP injection) occurs only during exploration. QAE is executed only after ITS resets, when the circuit is frozen. This ensures coherence and preserves the quadratic speedup. In [1] the error scales as $O(1/\sqrt{K})$.and in our QAE method the error scales as $O(1/K)$ which is an improvement over the orginal algorithm of [1]. The QAE algorithm estimates the mean from the angle between the 'solution' and 'non-solution' of from a qunatum state of the form used to initialse Grover's algorithm.

4 VARIANT 2 COHERENT REVERSIBLE EXPLORATION

Variant 2 extends Variant 1 by making exploration non-destructive.

4.1 Reversible Topological Perturbation

When ITS triggers a jitter operation:

1. Apply $U_{jitter}(\theta)$.
2. Optimize parameters on the perturbed landscape.
3. Once the loss decreases sufficiently, apply the inverse: $U^{-1}_{jitter}(\theta) = U_{jitter}(-\theta)$.

The circuit architecture returns to its original form, while parameters remain in the new region reached during exploration. This preserves the benefits of exploration while restoring a clean, stable circuit for QAE.

4.1.1 Adaptive Architecture Exploration

If the optimization trajectory becomes trapped in a barren plateau or a local minimum, the ITS metric triggers a recovery cycle to ensure global exploration:
Detection: If the ITS threshold is exceeded, the current parameterized unitary $U_{jitter}(\theta)$ is classified as a "dead-end," failing to effectively resolve the landscape's ill-posedness.
Unitary Inversion: Apply the inverse operator $U^{\dagger}_{jitter}(\theta)$ to the quantum circuit. This reverts the Born machine's topology to its original, stable base configuration.
Adaptive Branching: To avoid re-entering the failed configuration, the classical controller selects a new perturbation parameter $\theta' \notin \mathcal{H}$ (where $\mathcal{H}$ is the history buffer of previously attempted parameters).
Re-Perturbation: A new transformation $U_{jitter}(\theta')$ is applied. This maps the circuit to a new auxiliary landscape, creating a distinct probability distribution for the Born machine that avoids the previously encountered barren plateau.

To ensure efficient coverage, we propose two exploration strategies: a probabilistic approach relying on the measure-zero probability of repeating failed trajectories, and a deterministic strategy. The latter maintains a history buffer of applied unitary generators; upon stagnation, it triggers a unitary reversion to the degeneracy locus and selects a new generator from the available set, preventing redundant exploration and ensuring the algorithm systematically traverses the auxiliary manifold until it intersects the global basin of attraction

5 GENERALISED MACROSCOPIC SMOOTHING AND DYNAMIC SCHEDULING PROTOCOLS

The reversible topological perturbation described above can be generalised beyond a localized 2-qubit insertion to a macroscopic w-qubit deformation, establishing a stronger parallel to the broad smoothing radius of the classical Nash-Moser iteration. We define a locally gated operator $G_i(p_i, q_i) = I_1^{p_i} U_{\mathrm{pSWAP}}^{q_i}(\theta)$, where I_1 acts as a smoothed identity matrix. The exponent $q_i \in \{0, 1\}$ acts as a binary selector toggling the parameterized SWAP gate, while $p_i \in \mathbb{Z}$ is an integer that scales the depth of the smoothed identity perturbation. A generalized smoothing operator acting on a block of w qubits (where w is even) is constructed via the Kronecker tensor product of these local blocks:

$$U_{\text{macro}}(\theta) = \bigotimes_{i=1}^{w/2} G_i(p_i, q_i) \tag{100}$$

To govern the structural configuration of this macroscopic block when the "Iteration-Time-Stuck" (ITS) metric triggers, the exponents are driven dynamically as a function of the global iteration number t. Note that the ITS counter, here or in all of section 4, is strictly bounded from above by a stopping criterion, $ITS_{\max}$, to guarantee algorithmic termination. This upper bound is dynamically set based on the specific scenario, factoring in the complexity of the learning task and the acceptable computational deptNote that the ITS counter is strictly bounded from above by a stopping criterion, $ITS_{\max}$, to guarantee algorithmic termination (in success or failure). This upper bound is dynamically set based on the specific scenario, factoring in the complexity of the learning task and the acceptable computational depth. We introduce a dual-regime scheduling framework governed by a user-defined binary toggle switch $w_{\text{toggle}} \in \{0, 1\}$, allowing a choice between chaotic, exploratory pseudo-random behavior or deterministic linear pacing. The user can implement the w_{toggle} binary valuechange in various ways and w_{toggle} value is controlled by another subalgorithhm (this subagorithm to control w_{toggle} will be ITS triggered.. The simplest subalgorithm would just be to keep w_{toggle} a single fixed choce. Another choice would be to vary w_{toggle} dynamically, in a "periodic" style so its takes the values 0 and 1 more than unce when it it ITS trigerred more than once while the algoirthm is running:, so do $w_{\text{toggle}} = 1$ for some duration,according to some critera, so the algorithm does psudo random exploration for a minima, via equation 101, then so do $w_{\text{toggle}} = 0$ for some duration,according to some critera, so the algorithm do linear movement according to 102.

$$\chi_{\text{rand}}(t) = B^t \pmod{R} \tag{101}$$
$$\chi_{\text{lin}}(t) = \lfloor \alpha \cdot t \rfloor \tag{102}$$

where R is a large prime number acting as a modulus, B is a known primitive root modulo R ensuring a maximal-period pseudo-random sequence, and α is a classical scaling constant. There are more possiblities for $\chi_{\text{rand}}(t)$ and $\chi_{\text{lin}}(t)$ the equations 101 and 102 are examples. The active exponents are then determined via the convex-like toggle formulation. Because p_i accepts any integer, it takes the raw output of the schedule, whereas q_i is constrained to a binary state via a modulo-2 reduction:

$$p_i(t) = w_{\text{toggle}}(\chi_{\text{rand}}(t) - \chi_{\text{lin}}(t)) + \chi_{\text{lin}}(t) \tag{104}$$
$$q_i(t) = p_i(t) \pmod{2} \tag{103}$$

Crucially, the algebraic properties of the smoothed identity matrix ensure that $(I_1^{p_i})^{-1} = I_1^{-p_i}$. Because the inverse of a tensor product distributes cleanly over its individual matrix components-that is,

$$U_{\text{macro}}^{-1}(\theta) = \bigotimes_{i=1}^{w/2} (U_{\text{pSWAP}}^{-q_i}(\theta) I_1^{-p_i}) \tag{150}$$

-the coherent reversibility of the architecture remains flawlessly preserved regardless of the integer depth of p_i or the spatial size $2^w \times 2^w$ of the block. Upon escaping a barren plateau, applying $U_{\text{macro}}^{-1}(\theta)$ at the exact topological insertion site systematically annihilates the macroscopic perturbation ($U_{\text{macro}}^{-1} U_{\text{macro}} = I^{\otimes w}$), instantly restoring the entire w-qubit block to the static, coherent state required for subsequent Quantum Amplitude Estimation.

Remark 1 (The Commutative Limit and Global Topological Freedom): An important limiting case of the generalized scheduling protocol occurs when the binary selector is statically set to $q_i(t) = 0 \, \forall \, t$, collapsing the local operator to a pure power of the smoothed identity matrix, $G_i(p_i, 0) = I_1^{p_i}$. If I_1 is chosen to be the standard identity operator I (or a parameterized unitary constructed to commute globally with the generator Lie algebra of the hardware-efficient ansatz U_ϕ), the system enters a completely commutative regime where $[U_{\text{macro}}, U_\phi] = 0$.

Under this condition, the spatial and temporal constraints governing the application of the inverse operator are entirely relaxed. Because a globally commuting smoothing operator can traverse any arbitrary quantum gate without affecting its state transformation, $U_{\text{macro}}^{-1}(\theta)$ does not require localized topological adjacency to achieve cancellation. Instead, the inverse block can be applied across any arbitrary set of qubits or appended to the absolute terminus of the circuit timeline. It will mathematically commute through the optimized ansatz layers to achieve global annihilation ($U_{\text{macro}}^{-1} U_{\phi^*} U_{\text{macro}} = U_{\phi^*}$), offering total architectural flexibility during hardware execution.

6 ADVANTAGES OF THE PROPOSED METHODS

1. **Escaping Local Minima:** ITS-triggered pSWAP gates modify the entanglement structure to escape plateaus.

2. **Reversible Exploration:** Variant 2 ensures that exploration noise is fully coherent and reversible.

3. **Quadratic Speedup:** QAE reduces the number of Born-machine evaluations needed for gradients.

4. **Coherence Preservation:** Separating exploration (dynamic) and estimation (static) maintains the coherence required

   for QAE.

Theorem 1 (Quantum Nash–Moser Escape Principle)

Let $U(\theta)$ be a C^∞ one-parameter family of unitary operators acting on an n-qubit Born machine, satisfying:

- $U(0) = I$ (identity),
- $U(\theta)^\dagger = U(-\theta)$ (reversibility),
- $U(\theta)$ induces a smooth deformation of the circuit's entanglement topology (e.g., a parameterized SWAP).

Let $F(\phi)$ denote the variational free energy over parameters ϕ, and suppose gradient descent becomes trapped at a strict local minimum ϕ_t with:

$$\nabla_\phi F(\phi_t) = 0, \qquad \nabla_\phi^2 F(\phi_t) \succ 0. \tag{151}$$

Define the warped landscape:

$$F_\theta(\phi) = F(U(\theta) |\psi(\phi)\rangle). \tag{152}$$

Then:

(i) Smoothing

For sufficiently small $\theta \neq 0$,

$$\nabla_\phi F_\theta(\phi_t) \neq 0, \tag{153}$$

so the stationary point is removed and a descent direction is created.

(ii) Escape

Let ϕ_{t+k} be the parameters after k gradient steps on F_θ. Then generically

$$\phi_{t+k} \notin B(\phi_t), \tag{154}$$

where $B(\phi_t)$ is the basin of attraction of the original minimum.

(iii) Desmoothing

Applying the inverse unitary $U(-\theta)$ restores the original landscape:

$$F_{-\theta}(\phi) = F(\phi), \tag{155}$$

but does not return the parameters to the trapped point:

$$\phi_{t+k} \neq \phi_t. \tag{156}$$

(iv) Coherence Restoration

After desmoothing, the circuit is again static, enabling Quantum Amplitude Estimation to operate with Heisenberg-scaling precision.

(v) Global-Optimum Guarantee (by inheritance). Under the identification $V_{\text{geom}} \leftrightarrow iV_{\text{swap}}$ described in Appendix E.5, and provided that the pSWAP generators and ansatz generators satisfy assumptions (A1)–(A5) and (P1)–(P5) of the companion geometric paper under this translation, the corrected Global Convergence Theorem there applies verbatim. In particular, if the meta-learning algorithm maintains the incumbent

$$\hat{\phi}_t := \arg\min_{s \le t} F(\phi_s) \tag{500}$$

over all parameters ϕ_τ visited across ITS-triggered pSWAP episodes, then

$$\left\|\hat{\phi}_T - \phi^*\right\| \le \varepsilon \tag{501}$$

in time $T(\varepsilon)$ or $T_{\exp}(\varepsilon)$ (Appendix E or F of the companion paper [2], respectively, depending on which regime the pSWAP-generator DLA satisfies), with the complexity bound inherited from the geometric theorem.

Patience calibration for ITS (companion to Theorem 1). The non-improvement window s used by the ITS counter is chosen large enough that, once the warped landscape F_θ has entered a strongly-convex neighborhood of the true optimum, gradient descent reaches ε-accuracy *before* s consecutive non-improving steps elapse (see (P4′) of the companion geometric paper [2] for the precise bound). This guarantees ITS cannot trigger a jitter/branch away from a successful configuration before it has already produced an ε-accurate point.

Incumbent tracking. The meta-training loop maintains $\hat{\phi}_t := \arg\min_{s\le t} F(\phi_s)$ over all parameters visited, and reports $\hat{\phi}_T$ rather than the final ϕ_T.

Addition to Theorem 1. *(vi) Robust termination.* Even if ITS re-triggers after F_θ has entered the basin of the global optimum, the incumbent $\hat{\phi}_T$ retains the already-visited ε-accurate point, so conclusions (i)–(iv) survive repeated ITS activation.

In this paper, "global optimum" is used as shorthand for either (i) the true global minimizer or (ii) any point that satisfies the stopping tolerance criterion. The algorithm does not attempt to distinguish between these cases.

Interpretation

The unitary deformation $U(\theta)$ acts as a quantum smoothing operator, enabling escape from local minima; the inverse $U(-\theta)$ restores the original landscape without erasing progress. This constitutes a reversible, quantum-native analogue of the Nash–Moser iteration.

Related Work (Nash–Moser ↔ Quantum Optimization)

The classical Nash–Moser iteration was developed to address nonlinear problems where Newton-type methods fail due to loss of derivatives or rough functional landscapes. Its defining structure, graded smoothing, solution of the smoothed problem, and subsequent desmoothing while retaining progress, has since appeared in diverse areas of analysis and geometry. The nash Moser iteration is used an existence of solution in a global implicit function theorem. Although traditionally implemented using Fourier cutoffs or mollifiers, the core idea is independent of the specific smoothing operator.

In quantum optimization, adaptive ansatz methods such as ADAPT-VQE and quantum neural architecture search modify circuit structure to improve expressivity, but they do not incorporate reversible smoothing–desmoothing cycles. The dynamic pSWAP mechanism introduced in this work provides a quantum-native analogue of Nash–Moser smoothing: a smooth, invertible unitary homotopy ("mssing derivatives mention above in our context") that regularizes the variational landscape, enables escape from local minima, and is subsequently undone to restore coherence for amplitude estimation. This constitutes the first application of Nash–Moser–type reasoning in variational quantum meta-learning.

Diagram: Classical Nash–Moser vs Quantum pSWAP Mechanism

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                Classical Nash Moser vs Quantum pSWAP Mechanism
                ------------------------------------------------

   Classical Setting                                Quantum Setting
   -----------------                                ----------------

   Smoothing operator S$\sigma $                            Unitary smoothing U($\theta $)
   (Fourier cutoff / mollifier)                     (parameterized SWAP)

          |                                                 |
          v                                                 v

   Solve smoothed problem                           Optimize warped landscape
   (Newton step on S$\sigma $ f)                            (gradient descent on F$\theta $)
```

```
       |                                           |
       v                                           v

Desmoothing S$\sigma $^{-1}                     Apply inverse U(-$\theta $)
(remove smoothing)                     (restore original circuit)

       |                                           |
       v                                           v

Improved iterate retained              Improved parameters retained
(x_{k+1} $\neq $ x_k)                              ($\Phi ${t+k} $=\Phi $(t))

       |                                           |
       v                                           v

Repeat with finer $\sigma $                      Repeat with ITS controlled $\theta $
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The diagram express the structural equivalnce of the two schemes.
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7 CONCLUSION

We introduced two variants of quantum-aided Bayesian meta-learning that enhance the Born-machine variational posterior through dynamic entanglement adaptation, reversible exploration, and quantum amplitude estimation. These mechanisms jointly improve optimization robustness and sample efficiency, offering a promising direction for scalable quantum variational inference in discrete parameter spaces.Our unitary Homotopy framework could be used in other contexts such as VQE or QAOA for examples.

We acknowledge that while our pSWAP-based 'deform-solve-revert' mechanism structurally mimics the Nash-Moser iteration, this paper focuses on the algorithmic utility of this cycle for Born machine optimization. We presented this as an initial heuristic bridge between nonlinear analysis and quantum variational meta-learning, opening a pathway for potential future research between our "Unitary Homotopy' framework in the context of specific manifold embeddings.

References

[1] I. Nikoloska and .O. Simeone,QUANTUM-AIDED META-LEARNING FOR BAYESIAN BINARY NEURAL NETWORKS VIA BORN MACHINES ArXiv, 5th April 2022

[2] Locate reference

8 APPENDIX A

In this appendix we describe one way that the NET is linked to our quantum algorithm-there are other links between the NET and our quantum algorthm, not discussed her, but we discuss one method below. While the classical Nash Embedding Theorem guarantees that a Riemannian manifold can be isometrically embedded into a higher-dimensional Euclidean space,a fundamental statement on the existence of global geometry,our framework operates on an "inverted or reverse " iterative paradigm. We treat the Barren Plateau as a 'crumpled' or degenerate metric manifold, inherently pathologically flattened and resistant to gradient-based optimization. Our Unitary Homotopy framework functions as a 'reverse' Nash-Moser iteration: it 'decrumples' this degenerate landscape by dynamically lifting the variational parameters into an extrinsic, higher-dimensional auxiliary space. Topologically, this inflation expands the local curvature into a configuration structurally equivalent to a wedge sum of hyperspheres, restoring the metric's non-degeneracy. Consequently, while our method is inspired by Nash-Moser theory, we invoke it not to prove global geometric existence, but to engineer local algorithmic stability, allowing the optimizer to traverse and escape plateaus that are otherwise impenetrable; it uses the ideas of the Nash Moser iteration(used to prove a global result) in a different quantum based context to prove a local result instead of a global result.

While the classical Nash Embedding Theorem addresses the isometric embedding of n-dimensional manifolds into finite-dimensional Euclidean spaces, the Nash-Moser iteration itself was developed to resolve the pathological 'loss of derivatives' inherent in infinite-dimensional functional analysis. Our 'Unitary Homotopy' framework operates in the critical regime between these two regimes: it addresses the finite-dimensional parameter space of a variational quantum circuit, yet it targets the scaling pathologies,specifically the exponential vanishing of gradients (Barren Plateaus),that emerge as the system approaches the thermodynamic limit. Thus, we demonstrate that the Barren Plateau is the finite-dimensional quantum analogue of the analytical 'loss of regularity' encountered in PDEs. By employing tame estimates to restore metric non-degeneracy, our framework bridges finite-dimensional optimization and the functional-analytic rigor of the Nash-Moser iteration, providing a scalable regularizer that remains robust as the dimension of the Hilbert space and the circuit parameter space approach the infinite limit.

First we need to define how the variables in the main body of text are linked to this appendix as follows.

Perturbation Scaling ($\theta \Rightarrow \epsilon$): In the main body's heuristic discussion, the scalar variable θ was used as a single, multi-purpose parameter representing the rotation angle and general "jitter intensity" of a local $pSWAP$ gate ($U_{\text{jitter}}(\theta)$). In this appendix, we mathematically disentangle this action to support arbitrary macroscopic smoothing operators ($U_{\text{macro}}(\theta)$). Here, θ is generalized into a multi-dimensional parameter vector $\theta \in \mathbb{R}^k$ governing the baseline phase-space of the macro-gates , while the scalar intensity or "radius" of the geometric smoothing is isolated as the parameter $\epsilon \geq 0$. The localized unitary $U(\theta)$

from the main text thus maps directly to the macroscopic exponential form $U_{\mathrm{macro}}(\theta) = e^{-i\epsilon V}$, where V is the generalized Hermitian generator satisfying our quantum tame estimate.

Desmoothing Mechanics ($F_{-\theta} \Rightarrow \epsilon \to 0$): The main text heuristically notes that applying an inverse shift $(-\theta)$ restores the original landscape $(F_{-\theta}(\phi) = F(\phi))$. Geometrically, this "desmoothing" or "unwarping" cycle is formalized in two equivalent ways depending on the hardware execution strategy: either as a linear reversion phase where the smoothing intensity parameter smoothly decays to zero $(\epsilon \to 0)$, or via the explicit application of the coherent component-wise inverse block $U_{\mathrm{macro}}^{-1}(\theta)$. Both operations achieve the identical geometric result of systematically collapsing the auxiliary metric scaffolding $\tilde{g}_{ij}(\phi, \theta)$ back to the baseline physical metric constraints $g_{ij}(\phi)$ without losing the optimization progress achieved by the parameter vector ϕ

Discussion: The Intermediate Target Geometry and Inverse Nash Flow

To establish a rigorous dictionary between our framework and the classical Nash Embedding Theorem (NET), we must redefine the exact geometric nature of the perturbed manifold. In classical NET, the iteration protocol begins with a smooth abstract Riemannian manifold (M, g) and applies a sequence of short embeddings that become progressively more "crumpled" to converge upon a fixed, predetermined target metric g within a flat Euclidean space $\mathbb{R}^N$. Our Unitary Homotopy framework executes an Inverse Nash Flow. The baseline variational quantum landscape begins a priori in a pathologically degenerate state; the Quantum Fisher Information Metric (QFIM) $g_{ij}(\phi)$ is exponentially flattened by the barren plateau, compressing the local coordinate patches into unnavigable states.

When the Iteration Time Stuck (ITS) metric triggers, the application of the macroscopic smoothing operator $U_{macro}(\theta)$ does not merely perturb the system—it shifts the objective to an explicit intermediate target metric $\tilde{g}_{ij}(\phi, \theta)$. Mathematically, the manifold with this metric tensor is the Riemannian manifold $(T^M, \tilde{g}_\theta)$, constructed as the exact pullback of the Fubini-Study metric from the projective Hilbert space $\mathbb{P}(\mathcal{H})$ via the modified embedding function:

$$\tilde{f}_\theta : T^M \to \mathbb{P}(\mathcal{H}), \quad \phi \mapsto |\tilde{\psi}(\phi, \theta)\rangle = U_\phi U_{macro}(\theta)|0\rangle \tag{A504}$$

This auxiliary manifold $(T^M, \tilde{g}_\theta)$ acts as a highly symmetric, non-degenerate topological scaffold—analogous to a higher-dimensional "bunch of balloons." Within this targeted intermediate geometry, the non-vanishing commutator $[V, A_i] \neq 0$ acts as our quantum tame estimate, injecting structural variance that inflates the vanishing elements of the metric tensor into a strictly positive-definite profile.

The optimizer is thus permitted to traverse a smooth, feature-rich intermediate target space where gradients are well-conditioned and contraction mapping theorems hold. Once the optimizer successfully bypasses the coordinate bottlenecks of the plateau, the linear reversion phase ($\epsilon \to 0$) systematically deflates this auxiliary scaffolding, crumbling the manifold back to its physical constraints (T^M, g_0) and depositing the parameter vector securely within the global minimum.

To fully appreciate the novelty of the Quantum Nash-Moser scheme, it is instructive to contrast its geometric trajectory with the classical NET. In classical NET, the iteration protocol introduces progressively “crumpled,” high-frequency metric deformations to force an isometric embedding. Our framework inverts this thermodynamic-like flow. The application of the macroscopic smoothing operator $U_{macro}(\theta)$ acts as a non-local embedding functor. Rather than smoothing a continuous function, it maps the degenerate parameter space into a highly symmetric, higher-dimensional auxiliary manifold—topologically analogous to a wedge sum of hyperspheres or a highly entangled fiber bundle. Within this targeted higher-dimensional scaffolding, the injected topological noise restores non-vanishing sectional curvatures. Consequently, the “target metric” of our iteration is shifted from a static terminal geometry to this temporary, curved auxiliary topology $\tilde{g}_{ij}(\phi, \theta)$. Once the optimizer bypasses the coordinate bottlenecks of the plateau, the systematic deflation of this auxiliary scaffolding ensures that the system settles into the global minimum of the target space, having utilized the higher-dimensional “ballooning” geometry purely as an optimization catalyst.

Resolving the "Loss of Derivatives" via Commutator Inequalities

To establish a strict mathematical dictionary between our framework and the classical Nash-Moser iteration, we must address the fundamental problem the classical theorem was built to solve: the “loss of derivatives.” In the classical context of solving non-linear partial differential equations, the inverse of the linearized operator lacks bounded regularity, meaning each Newton iteration step yields a solution that loses differentiability order. Nash tamed this loss of regularity not by assumption, but by introducing a smoothing operator backed by strict a priori inequalities (tame estimates) that bound the functional norms at each step.

In our variational quantum optimization paradigm, the phenomenon of a barren plateau represents the exact optimization analogue to this classical loss of derivatives. As the qubit count scales, the partial derivatives of the cost function flatline exponentially ($\partial_i E \to 0$), resulting in an asymptotic loss of effective derivative (gradient) information.

Rather than assuming a priori that a navigable auxiliary manifold exists, the Unitary Homotopy framework explicitly constructs it and guarantees its non-degeneracy via algebraic inequalities. By selecting a macroscopic smoothing generator V that does not commute with the ansatz’s local generators A_i, we enforce a strict lower bound on the metric perturbation tensor via the operator norm inequality:

$$\|[V, A_i]\|_F \geq c > 0 \tag{A505}$$

where $\|\cdot\|_F$ is the Frobenius norm and c is a non-vanishing structural constant. This inequality acts as our quantum "tame estimate." It mathematically guarantees that the first-order tracking tensor $h_{ij}^{(1)}(\theta)$ injects sufficient regularizing variance to lift the metric $\tilde{g}_{ij}$ out of the degenerate plateau. Consequently, the temporary target metric of our "bunch of balloons" geometry is proven to exist as a well-conditioned, strictly positive-definite Riemannian manifold. The quantum smoother effectively "adds the derivatives" back into the optimization space, mirroring Nash's use of mollifiers to restore differentiability to an otherwise unnavigable functional landscape.

Theorem A1 (Existence of the Non-Degenerate Auxiliary Manifold) Let $\mathcal{M}_0$ be a finite-dimensional parameter manifold of a variational quantum circuit residing in a barren plateau, such that its baseline Quantum Fisher Information Metric (QFIM) is highly degenerate, $g_{ij}(\phi) \to 0$. Let V be the Hermitian generator of a generalized macroscopic smoothing operator $U_{macro}(\theta) = e^{-i\epsilon V}$. If V satisfies the quantum tame estimate with respect to the ansatz's local generators A_i such that the commutator norm is strictly bounded from below, $\|[V, A_i]\| \geq c > 0$ for all i, then for a sufficiently small smoothing intensity $\epsilon > 0$, the perturbed metric tensor $\tilde{g}_{ij}(\phi, \theta)$ of the auxiliary manifold $\mathcal{M}_\theta$ is strictly positive-definite.

Proof Sketch:
To prove that the auxiliary manifold $\mathcal{M}_\theta$ exists as a strictly positive-definite Riemannian manifold, we must demonstrate that its metric tensor $\tilde{g}_{ij}$ is positive-definite. By definition, a metric tensor is positive-definite if and only if its associated quadratic form is strictly positive for any non-zero parameter velocity vector $\mathbf{v} \in \mathbb{R}^M$. That is, we require:

$$Q(\mathbf{v}) = \sum_{i,j} v_i \tilde{g}_{ij} v_j > 0 \quad \text{for all } \mathbf{v} \neq \mathbf{0} \tag{A506}$$

We begin by expressing the perturbed metric tensor as the covariance matrix of the transformed local generators in the Heisenberg picture. The smoothed generator is given by $\tilde{A}_i = e^{i\epsilon V} A_i e^{-i\epsilon V}$. For an arbitrary vector $\mathbf{v}$, we define the directional operator:

$$\tilde{A}(\mathbf{v}) = \sum_i v_i \tilde{A}_i = \sum_i v_i \Big(A_i + i\epsilon[V, A_i] + \mathcal{O}(\epsilon^2) \Big) \tag{A507}$$

The quadratic form $Q(\mathbf{v})$ is exactly the variance of this directional operator evaluated with respect to the initial state $|0\rangle$:

$$Q(\mathbf{v}) = \mathrm{Var}[\tilde{A}(\mathbf{v})] = \langle 0|\tilde{A}(\mathbf{v})^2|0\rangle - \langle 0|\tilde{A}(\mathbf{v})|0\rangle^2 \tag{A508}$$

In the strict barren plateau regime, the variance of the unperturbed baseline operator approaches zero ($\mathrm{Var}[\sum v_i A_i] \approx 0$). Therefore, substituting the BCH expansion of $\tilde{A}(\mathbf{v})$ into the variance isolates the geometric contribution of the smoother:

$$Q(\mathbf{v}) \approx \epsilon^2 \mathrm{Var}\left[\sum_i v_i (i[V, A_i])\right] \tag{A509}$$

(Note: Depending on the symmetry of the initial state, the first-order $\mathcal{O}(\epsilon)$ cross-terms may vanish, making the second-order variance of the commutator the leading strictly positive term).
By our initial hypothesis, the generalized smoother V is deliberately chosen such that it structurally does not commute with the ansatz layers, satisfying the tame estimate $\|[V, A_i]\| \geq c > 0$. Because the commutator is non-vanishing and Hermitian (up to a phase i), the operator $i[V, A_i]$ represents a valid, non-trivial observable.
Consequently, for any non-zero vector $\mathbf{v}$, the variance of this linearly combined commutator operator is bounded away from zero:

$$\mathrm{Var}\left[\sum_i v_i (i[V, A_i])\right] \geq K > 0 \tag{A510}$$

where K is a positive constant dependent on the structural lower bound c.
Therefore, the quadratic form of the perturbed metric yields:

$$Q(\mathbf{v}) \geq \epsilon^2 K > 0 \tag{A511}$$

Because $Q(\mathbf{v}) > 0$ for all non-zero $\mathbf{v}$, the perturbed metric tensor $\tilde{g}_{ij}$ is strictly positive-definite. This mathematically guarantees that the generalized smoothing operator V successfully lifts the degenerate barren plateau geometry into a curved, non-degenerate auxiliary manifold (the topological "bunch of balloons"), allowing standard gradient-based optimizers to reliably traverse the space. ■

Theorem A2 (Convergence via Quantum Contraction Mapping)
Let $\mathcal{M}_\theta$ be the auxiliary manifold equipped with the non-degenerate perturbed metric tensor $\tilde{g}_{ij}(\phi, \theta)$ established in Theorem 1. Let $T : \mathcal{M}_\theta \to \mathcal{M}_\theta$ defined by $T(\phi) = \phi - \eta \tilde{g}^{-1}(\phi) \nabla E(\phi)$ represent the natural gradient optimization operator. For a sufficiently tuned learning rate $\eta > 0$, the operator T is a strict contraction mapping under the Riemannian distance metric, ensuring sequence convergence to a unique fixed-point solution ϕ^*.

Proof (Sketch):
To prove that the quantum optimization sequence $\phi_{k+1} = T(\phi_k)$ converges to a unique fixed point, we must demonstrate that T satisfies the Banach Contraction Principle within a local coordinate patch $\mathcal{U} \subset \mathcal{M}_\theta$.
We evaluate the Jacobian of the optimization mapping $T(\phi)$:

$$J_T(\phi) = I - \eta \tilde{g}^{-1}(\phi) \mathcal{H}_E(\phi) - \eta \left(\partial_\phi \tilde{g}^{-1}(\phi)\right) \nabla E(\phi) \tag{A512}$$

where $\mathcal{H}_E(\phi)$ is the Hessian matrix of the cost function $E(\phi)$. Near the critical regions of the barren plateau, the gradient vector is small ($\nabla E(\phi) \to 0$), allowing us to drop the third term and focus on the primary structural operator:

$$J_T(\phi) \approx I - \eta \tilde{g}^{-1}(\phi)\mathcal{H}_E(\phi) \tag{A513}$$

By Theorem 1, the generalized smoother $U_{macro}(\theta)$ ensures that the auxiliary metric $\tilde{g}_{ij}$ is strictly positive-definite and bounded, implying its eigenvalues are bounded: $0 < \lambda_{min} I \leq \tilde{g} \leq \lambda_{max} I$. Furthermore, on this smoothed manifold, we assume the cost function Hessian exhibits bounded spectral regularity within the local convex basin, satisfying $\alpha I \leq \mathcal{H}_E \leq \beta I$, where $\alpha > 0$ represents the minimum curvature.

The operator T is a strict contraction if the operator norm of its Jacobian is strictly bounded by unity: $\|J_T(\phi)\| \leq L < 1$. Substituting our spectral bounds:

$$\|I - \eta \tilde{g}^{-1}\mathcal{H}_E\| \leq \max\left(\left|1 - \eta \frac{\alpha}{\lambda_{max}}\right|, \left|1 - \eta \frac{\beta}{\lambda_{min}}\right| \right) \tag{A514}$$

By selecting a learning rate η that strictly satisfies:

$$0 < \eta < \frac{2\lambda_{min}}{\beta} \tag{A515}$$

the operator norm is forced strictly below 1 ($L < 1$).

Therefore, by the Banach Contraction Mapping Theorem, the natural gradient flow over the auxiliary "bunch of balloons" manifold forms a strict contraction. This guarantees that the optimization sequence converges to a unique fixed point ϕ^* within the patch, theoretically proving that the injected geometric curvature provides a stable, deterministic escape vector out of the barren plateau before the algorithm execution reverts ($\epsilon \to 0$) to the baseline manifold constraints. ■

9 APPENDIX B

Global Reachability and Geometric Justification.Proof of Global Convegence (under reasonable assumptions)

Recall when we mean global convergence we mean reaching a stopping criteria (as is ususal in neural networks) which may also be mean it stops and finds the global optima but need not be the global optima. The reversible pSWAP-based "deform-solve-revert" mechanism described in this work admits a precise geometric formulation within the finite-dimensional Nash-Moser analogue developed in our companion paper which is reference [2]. We will give this proof in out future paper [2]. We refer the reader to [2] for the proof of global convergence, on reversible smoothing for degenerate Riemannian metrics. The global convergence above is conditionalon the following.

Scope of this Justification. As in Section 6, this justification is conditional: it holds once (A1)–(A5) and (P1)–(P5) of [2] are verified for the specific pSWAP generators and ansatz $\{A_i\}$ in use, not automatically from the V-correspondence of Appendix E.5 alone.
In that setting, the Born-machine parameter manifold M is embedded into an ambient Riemannian space (N, h), and the pSWAP-driven unitary homotopy is realized as a Lie-group action Φ_ϵ that inflates the degenerate pullback metric into a strictly positive-definite auxiliary metric on a family of "balloons" attached to the barren-plateau locus.
Under a commutator-based tame estimate

$$\|[V, A_i]\| \geq c > 0 \tag{B600}$$

, an orbit-transversality condition for the Lie action, and the assumption that the global minimizer of the variational free energy is a unique global attractor, the resulting Lie-orbit branching induces a star-graph topology of auxiliary manifolds emanating from the degeneracy locus. As shown in the geometric theorem, this structure precludes isolated traps: every time the optimization dynamics becomes stuck on a barren plateau, the reversible inflation–deflation cycle returns the state to the hub and branches it into a new auxiliary manifold. Because the bouquet of balloons covers the search space, this process is guaranteed (in the asymptotic-time sense) to enter the basin of the global minimum, providing a differential-geometric justification of our "Quantum Nash–Moser escape principle." ,See]2] for futher details.

Appendix C.

Lemma (Generic Tame Generator in Finite-Dimensional Quantum Lie Algebras)
Let $\mathfrak{g}$ be a finite-dimensional Lie algebra with trivial center (e.g., $\mathfrak{su}(2^n)$). Let $\{A_1, \ldots, A_m\} \subset \mathfrak{g}$ be a fixed finite set of nonzero generators. Let V be drawn from any absolutely continuous probability distribution on $\mathfrak{g}$ (e.g., Gaussian coefficients in any basis). Define the commutator norms

$$X_i(V) := \|[V, A_i]\|_F. \tag{C700}$$

(1) With probability 1 (in a "measure sense"), all commutators are nonzero. Because $\mathfrak{g}$ has a trivial center and $A_i \neq 0$, the centralizer

$$\mathrm{Cent}(A_i) := \{V \in \mathfrak{g} : [V, A_i] = 0\} \tag{C701}$$

is a proper linear subspace of $\mathfrak{g}$. Every proper subspace of a finite-dimensional vector space has Lebesgue measure zero. Since a finite union of measure-zero sets is measure zero,

$$\Pr([V, A_i] \neq 0 \quad \forall i) = 1. \tag{C702}$$

(2) With probability 1 , the minimum commutator norm is strictly positive. Each $X_i(V)$ is a continuous, nonzero random variable almost surely. Since the index set is finite, the minimum

$$c(V) := \min_{1 \leq i \leq m} X_i(V) \tag{C703}$$

is strictly positive almost surely.
Lemma Conclusion.
With probability 1 , a randomly chosen V satisfies the quantum tame estimate

$$\|[V, A_i]\|_F \geq c(V) > 0 \quad \forall i. \tag{C704}$$

Thus the tame condition required for the smoothing and escape theorems is a generic property, satisfied by almost every choice of smoothing generator V.

APPENDIX D

Extension of the Unitary Homotopy Framework to General Variational Quantum Algorithms (VQAs)

This appendix extends the reversible Unitary Homotopy framework beyond Born machines to the full class of variational quantum algorithms (VQAs), including VQE and QAOA. The key observation is that all VQAs share the same structural template:

$$U(\theta) = U_L(\theta_L) \cdots U_1(\theta_1), \quad C(\theta) = \langle 0 \mid U^\dagger(\theta) O U(\theta) \mid 0 \rangle, \tag{D800}$$

optimized via gradient descent.
Barren plateaus correspond to vanishing directional derivatives, i.e., degeneracy of the Quantum Fisher Information Metric (QFIM). The Unitary Homotopy mechanism resolves this degeneracy exactly as in the Born-machine case.

D.1 Generalized Warped Landscape for Arbitrary VQAs

For any VQA cost function $C(\theta)$, define the warped landscape:

$$C_\epsilon(\theta) = C(U(\epsilon)U(\theta)), \tag{D801}$$

where $U(\epsilon) = \exp(\epsilon V)$ is a reversible Lie-group deformation.
The gradient transforms as:

$$\nabla_\theta C_\epsilon(\theta) = \nabla_\theta C(\theta) - \epsilon\, E_V(\theta) + \mathcal{O}(\epsilon^2), \tag{D802}$$

where $E_V(\theta)$ is the commutator-induced curvature injection term.
Thus, even when $\nabla_\theta C(\theta) = 0$, the warped landscape satisfies:

$$\nabla_\theta C_\epsilon(\theta) \neq 0, \tag{D803}$$

providing an escape direction.

D.2 Explicit Non-Commutativity Requirement (Quantum Tame Estimate)

To guarantee curvature injection, the smoothing generator V must be chosen so that it does not commute with the local generators A_j of the ansatz:

$$[V, A_j] \neq 0 \quad \text{for all relevant } j. \tag{D804}$$

This is the finite-dimensional analogue of the Nash–Moser tame estimate:

$$\|[V, A_j] \cdot p_0\| \geq c > 0, \tag{D805}$$

ensuring that the metric perturbation tensor is bounded away from zero.
This requirement is essential for:

- **VQE** (hardware-efficient ansätze often have large commutative subalgebras)
- **QAOA** (alternating unitaries generate structured Lie algebras with symmetries)

Choosing V outside the centralizer of the ansatz generators ensures that the deformation injects curvature into all degenerate directions.

D.3 Coherence Restoration for Quantum Amplitude Estimation (QAE)

QAE requires:

- a static,
- fully coherent,
- time-independent circuit.

During exploration, the deformation $U(\epsilon)$ modifies the circuit topology. Before QAE is executed, the inverse deformation $U(-\epsilon)$ is applied:

$$U(-\epsilon)U(\epsilon) = I, \tag{D806}$$

which annihilates the macroscopic perturbation exactly, restoring:

- full coherence,
- the original circuit architecture,
- the static unitary required for QAE's Heisenberg-scaling precision.

This is the quantum analogue of Nash–Moser desmoothing.

D.4 Specialization to VQE

For VQE, the cost is:

$$C(\theta) = \langle \psi(\theta) \mid H \mid \psi(\theta) \rangle. \tag{D807}$$

Barren plateaus arise from:

- deep hardware-efficient ansätze
- global observables
- noise-induced metric collapse

The reversible deformation injects curvature into the QFIM:

$$g_{ij}(\theta) \to g_{ij}^{(\epsilon)}(\theta) \tag{D808}$$

and the inverse deformation restores coherence for measurement or QAE.

D.5 Specialization to QAOA

QAOA uses alternating unitaries:

$$U(\gamma, \beta) = e^{-i\beta B} e^{-i\gamma C}. \tag{D809}$$

Symmetry sectors and commutative subalgebras induce barren plateaus.
Choosing V such that:

$$[V, B] \neq 0, \quad [V, C] \neq 0, \tag{D810}$$

ensures curvature injection. The reversible deformation perturbs the QAOA landscape without destroying its structure, and the inverse deformation restores the original circuit.

D.6 Summary

The Unitary Homotopy framework applies to all VQAs because:

- all VQAs induce a QFIM
- barren plateaus correspond to metric degeneracy
- non-commuting Lie-group smoothing injects curvature
- the deformation is reversible
- QAE coherence is fully restored by $U(-\epsilon)$

Born machines are simply one specialisation of this general mechanism.

Appendix E . Continuous Two-Qubit pSWAP, Hermitian Generator, Adjoint Expansion, and Lie-Algebra Correspondence

This appendix provides the exact continuous parameter formulation of the two-qubit pSWAP gate used in the main algorithm. We derive its Hermitian generator, compute the adjoint expansion with the appropriate sign, and state explicitly how this unitary,group Lie algebra corresponds to the real Lie algebra used in the geometric paper [2]. This establishes the precise mathematical bridge between the reversible smoothing framework and the pSWAP-based unitary homotopy.

E.1 Two-Qubit pSWAP Gate

In the computational basis

$$\{|00\rangle,\ |01\rangle,\ |10\rangle,\ |11\rangle\},$$

the parameterized SWAP (pSWAP) gate performs a rotation in the $\{|01\rangle, |10\rangle\}$ subspace while leaving $|00\rangle$ and $|11\rangle$ invariant:

$$\mathrm{pSWAP}(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{E900}$$

This is a two qubit gate and it cannot be written as a tensor product of single-qubit gates, because it mixes the amplitudes of $|01\rangle$ and $|10\rangle$, producing non-local behavior essential for entanglement adaptation.

E.2 Hermitian Generator V_{swap}

We extract the Hermitian generator V_{swap} such that

$$\mathrm{pSWAP}(\theta) = \exp(-i\theta V_{\mathrm{swap}}). \tag{E901}$$

Differentiating at $\theta = 0$:

$$\frac{d}{d\theta}\mathrm{pSWAP}(\theta)\Big|_{\theta=0} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{E902}$$

Since

$$\frac{d}{d\theta}\exp(-i\theta V)\Big|_{\theta=0} = -iV,$$

we obtain:

$$V_{\mathrm{swap}} = i \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{E903}$$

This generator is Hermitian, and $\exp(-i\theta V_{\mathrm{swap}})$ reproduces the full pSWAP gate exactly for all θ.

E.3 Adjoint Expansion

Define the one–parameter unitary flow:

$$U(\varepsilon) = \exp(-i\varepsilon V_{\mathrm{swap}}). \tag{E904}$$

For any operator A acting on the same two–qubit space, the adjoint action is:

$$A_\varepsilon = U(-\varepsilon)\, A\, U(\varepsilon). \tag{E905}$$

Using the Baker-Campbell-Hausdorff expansion for unitary flows:

$$A_\varepsilon = A + i\varepsilon[V_{\mathrm{swap}}, A] + O(\varepsilon^2). \tag{E906}$$

E.4 Embedding into the Full Circuit

In the n-qubit Born machine, the jitter operator is:

$$U_{\mathrm{jitter}}(\theta) = I^{\otimes(i-1)} \otimes \mathrm{pSWAP}_{i,i+1}(\theta) \otimes I^{\otimes(n-i-1)}. \tag{E907}$$

Its generator is:

$$V_{\mathrm{swap}}^{(i)} = I^{\otimes(i-1)} \otimes V_{\mathrm{swap}} \otimes I^{\otimes(n-i-1)}. \tag{E908}$$

The adjoint expansion becomes:

$$U_{\mathrm{jitter}}(-\varepsilon)\, A\, U_{\mathrm{jitter}}(\varepsilon) = A + i\varepsilon[V_{\mathrm{swap}}^{(i)}, A] + O(\varepsilon^2). \tag{E909}$$

This is the exact two-qubit analogue of the geometric adjoint expansion used in the reversible smoothing framework.

E.5 Lie-Algebra Correspondence

Our differential geometry paper [2] with a real Lie algebra and uses the expansion:

$$\mathrm{Ad}_{\exp(\varepsilon V_{\mathrm{geom}})}(A) = A - \varepsilon[V_{\mathrm{geom}}, A] + O(\varepsilon^2). \tag{E910}$$

The unitary group uses anti-Hermitian generators:

$$X = -iV_{\mathrm{swap}}, \tag{E911}$$

$$X = -i\,V_{\mathrm{swap}}, \qquad X^{\dagger} = -X. \tag{E912}$$

Under the identification

$$V_{\mathrm{geom}} \;\longleftrightarrow\; X = -iV_{\mathrm{swap}}, \tag{E913}$$

the two adjoint expansions coincide:

$$U(-\varepsilon)AU(\varepsilon) = A + i\varepsilon[V_{\mathrm{swap}}, A] + O(\varepsilon^2), \tag{E914}$$

which is the unitary-group analogue of

$$A - \varepsilon[V_{\mathrm{geom}}, A] + O(\varepsilon^2). \tag{E915}$$

This explicit identification is an "exact correspondence" between the two papers/triology lets discuss in detail in the next section what this means more precisely.

Inheritance of Geometric Assumptions (Conditional). Under the Lie-algebra identification established in this appendix, the pSWAP generator iV_{swap} corresponds exactly to the geometric smoothing generator V_{geom}, and the associated adjoint expansions coincide. This identification transfers only the *smoothing operator itself* – it does *not* establish the surrounding structural hypotheses. Assumptions (A1)–(A5) and (P1)–(P5) also depend on the ansatz generators $\{A_i\}$, the dimension and coverage properties of the resulting dynamical Lie algebra, and the tame estimate

$$\|[V, A_i] \cdot p_0\|_h \;\geq\; c\|v\| \tag{E916}$$

evaluated at the physical input state p_0 – none of which follow from the V-correspondence alone. We therefore do *not* claim verbatim inheritance; rather, (A1)–(A5) and (P1)–(P5) must be separately verified for the specific hardware-efficient ansatz $\{A_i\}$ and DLA in use, with V_{geom} replaced by iV_{swap} throughout. Clause (v) of Theorem 1 and the Global Convergence Theorem of the companion paper apply only once this verification is carried out for the ansatz in question.

E.6 Multi-Parameter Tensor-Product Deformations

The geometric paper's multi0parameter reversible smoothing uses a family of generators

$$V_1, \ldots, V_m \tag{E917}$$

and a multi–parameter flow

$$U(\varepsilon_1, \ldots, \varepsilon_m) = \exp(\varepsilon_1 V_1) \cdots \exp(\varepsilon_m V_m). \tag{E918}$$

The quantum algorithm implements the exact analogue using multiple pSWAP gates inserted at different qubit locations:

$$U_{\text{jitter}}(\vec{\theta}) = \prod_{k=1}^{m} \Big(I^{\otimes(i_k-1)} \otimes \text{pSWAP}_{i_k, i_k+1}(\theta_k) \otimes I^{\otimes(n-i_k-1)} \Big). \tag{E919}$$

Each pSWAP has generator

$$V_{\text{swap}}^{(i_k)} = I^{\otimes(i_k-1)} \otimes V_{\text{swap}} \otimes I^{\otimes(n-i_k-1)}. \tag{E920}$$

Thus the multi-parameter unitary flow is

$$U(\vec{\varepsilon}) = \exp(-i\varepsilon_1 V_{\text{swap}}^{(i_1)}) \cdots \exp(-i\varepsilon_m V_{\text{swap}}^{(i_m)}). \tag{E921}$$

The adjoint expansion becomes

$$U(-\vec{\varepsilon})\, A\, U(\vec{\varepsilon}) = A + i \sum_{k=1}^{m} \varepsilon_k\, [V_{\text{swap}}^{(i_k)}, A] + O(\|\vec{\varepsilon}\|^2). \tag{E922}$$

This is the exact unitary-group analogue of the geometric paper's multi-parameter curvature-injection formula

$$A_\varepsilon = A - \sum_{k=1}^{m} \varepsilon_k\, [V_k, A] + O(\|\varepsilon\|^2). \tag{E923}$$

The sign and i-factor are accounted for by the identification in Section E.5. Thus the tensor-product structure, the multi-parameter reversible smoothing, and the curvature-injection mechanism match our geometric paper [2] exactly .

Macroscopic Scheduling. The continuous pSWAP operator introduced in this appendix, $U_{\text{pSWAP}}(\varepsilon) = e^{-i\varepsilon V_{\text{swap}}}$, provides the local Lie-group deformation used in the reversible smoothing framework. This operator is parameterized solely by the continuous smoothing intensity ε. In contrast, the macroscopic operator defined in the main text employs additional discrete scheduling parameters p_i and q_i, which control the depth and activation of each local block. Hence we write each block explicitly as

$$G_i(p_i, q_i; \varepsilon) = I^{\,p_i} \big(U_{\text{pSWAP}}(\varepsilon)\big)^{q_i}, \tag{E924}$$

where p_i and q_i are discrete control indices, while ε remains the continuous Lie-group parameter. The macroscopic smoothing operator is then

$$U_{\text{macro}}(\varepsilon) = \bigotimes_{i=1}^{w/2} G_i(p_i, q_i; \varepsilon). \tag{E925}$$

In other words Appendix E specifies the continuous local deformation, while the main text specifies the higher-level scheduling protocol built from these blocks.

Inheritance of Geometric Assumptions. Because the pSWAP generator iV_{swap} is identical, under the Lie-algebra identification established in Appendix E.5, to the geometric smoothing generator V_{geom} used in the companion paper, all structural assumptions (A1)–(A5) and (P1)–(P5) verified there for V_{geom} apply verbatim to the pSWAP-driven architecture.

E.6 Exactness of geometric Correspondence.

Using the resuts above and In the next paper will show that the PSWAP deformation can be exactly deform the Born machine landscape in such a way that it is an exact way not an approximation that can be represented from a rotated base point within the same ambient projective Hilbert space which is exactly the same as the Born's machine search landscape.

E.7 Summary of the Appendix

This appendix establishes the following:

- The pSWAP operator is replaced by its continuous version of the SO(2) Lie-group analogue.
- All smoothing operations become smooth one-parameter flows, compatible with the geometric Nash–Moser theory.
- The macroscopic smoothing operator is rewritten as a continuous tensor-product Lie-group operator.
- All convergence theorems in the geometric companion paper now rigorously apply to the quantum algorithm.
- Multi-parameter SWAP smoothing operators discussed.